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Enrichment of carbon dioxide using Mg–Al layered double hydroxides

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ABSTRACT

Owing to the low atmospheric concentration of CO₂ (400 ppm), direct air capture technologies require adsorbents that offer high CO₂ adsorption capacities at low CO₂ concentrations. In this study, we investigated the potential of CO₃Mg–Al layered double hydroxides (LDHs; Mg/Al molar ratio = 2, 3, and 4) as CO₂ adsorbents. The CO₂ adsorption and desorption behavior was studied at various molar ratios, adsorption/desorption temperatures, and linear speeds. CO₂ adsorption by CO₃Mg–Al LDH followed the Langmuir model, indicating that monolayer adsorption occurs. The maximum CO₂ adsorption quantity calculated using the Langmuir equation was 0.794 mmol/g for CO₃Mg–Al LDH with a Mg/Al molar ratio of 3. In dry CO₂ adsorption experiments, a maximum CO₂ adsorption quantity of 0.319 mmol/g was recorded with a Mg/Al molar ratio of 3 and temperature of 60 °C. In dry CO₂ desorption experiments, the CO₂ concentration and desorption quantity reached maximum values at 300 °C and a linear speed of 0.250 m/min. Notably, the maximum CO₂ concentration was 6232 ppm, showing a 15.6-fold increase in comparison with the initial concentration of 400 ppm. Overall, our results demonstrate that CO₃Mg–Al LDH has high potential as a CO₂ adsorbent. The study findings have significant implications for the development of new CO₂ adsorbents for the direct air capture and storage of CO₂.

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1. Introduction

CO₂ capture and utilization technologies aim to separate and collect CO₂ from the atmosphere or exhaust gases, and then use it as a resource for crop fertilization or chemical production. Because CO₂ capture from exhaust gases is restricted to facilities adjacent to plants discharging exhaust gas, such as thermal power plants, the focus of this study is on the direct air capture of CO₂ from the atmosphere. Given that the atmospheric concentration of CO₂ is approximately 400 ppm, it is crucial that the adsorbents used in this process have a high CO₂ adsorption capacity at low CO₂

concentrations, and that the reaction system can handle large volumes of gas.

Layered double hydroxides (LDHs) are a class of materials composed of positively charged layers of divalent metal ions and negatively charged layers of anions. The structure of an LDH is composed of (i) a host layer, which has a secondary layered structure with a regular octahedral surface covered with positive charges due to substitution of some of the divalent metal hydroxides (M(OH)₂) with trivalent metal ions; and (ii) a guest layer, which is composed of anions that compensate for the positive charges of the host layer, and intercalated H₂O (Cavani et al., 1991; Mills et al., 2012). The general formula for an LDH is expressed as

$$[M^{2+}_{1-x}M^{3+}_x(OH)_2][A^{n-}_{(x/n)} \cdot mH_2O] \quad (1)$$

where M²⁺ is a divalent metal ion, M³⁺ is a trivalent metal ion, Aⁿ⁻ is an n-valent anion, x is the ratio of trivalent metal ions

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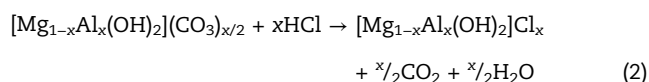
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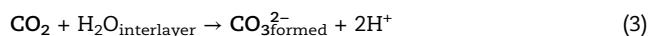
to the total number of metal ions ($x = M^{3+}/(M^{2+} + M^{3+})$), and $m = 1 - (3x/2)$ (Miyata, 1980).

In the host layer of an LDH, divalent metal ions such as Mg, Zn, Co, Ni, and Fe and trivalent metal ions such as Al, Fe, Cr, In, and Co can be used to alter the functionality of the material. The trivalent metal ions of the host layer have a solid-solubility limit, x , which is determined by Vegard's rule, and is typically between 0.20 and 0.33 (Miyata, 1980). The guest layer of LDHs can be intercalated with various types of anions, which introduces a wide range of potential applications as functional materials (Kameda et al., 2019a, 2021a, 2021b; Zhu et al., 2022). The greater the charge density of the intercalated anions, the more readily they are incorporated. When LDHs are heated, the intercalated H_2O is released, and continued heating can lead to the breakdown and oxidation of the host layer (Miyata, 1980; Sato et al., 1986; Kameda et al., 2015). Mg - Al LDHs intercalated with CO_3^{2-} (CO_3 - Mg - Al LDH) are often used as precursors for the synthesis of oxides. The release of intercalated H_2O occurs at temperatures between 100 and 250 °C, and the breakdown of the host layer occurs at up to 400 °C. Therefore, the layered structure can be maintained after the intercalated H_2O is eliminated. However, once the host layer breaks down at around 400 °C, the layered structure is completely lost and Al forms a solid solution in MgO , forming a Mg - Al layered double oxide with an $NaCl$ -type structure and cation defects. If heating is continued, this material breaks down further to form MgO and spinel ($MgAl_2O_4$) at around 800 °C.

We previously investigated the treatment of acid gases, such as HCl , SO_2 , and NO_2 , using a dry method with CO_3 - Mg - Al LDH (Kameda et al., 2019b, 2019c, 2020a, 2020b, 2020d; Takahashi et al., 2021). We also found that the acid gas removal rate decreases in the presence of coexisting CO_2 (Kameda et al., 2020c). For HCl , the removal of HCl by CO_3 - Mg - Al LDH is attributed to the reaction between CO_3^{2-} in the interlayer of Mg - Al LDH (i.e., intercalated CO_3^{2-}) and HCl , according to Eq. (2), followed by the intercalation of Cl^- in the interlayer of Mg - Al LDH (Kameda et al., 2020b). The reaction between interlayer CO_3^{2-} and HCl (Eq. (2)) is classified as chemisorption.

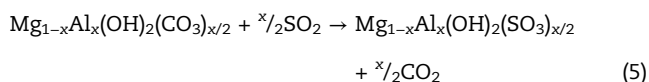


There is an obvious decrease in the HCl removal ability upon introducing 5 % CO_2 . This is ascribed to the exchange reaction between intercalated CO_3^{2-} and gaseous CO_2 , which inhibits the exchange reaction between intercalated CO_3^{2-} and HCl . Gaseous CO_2 becomes intercalated in Mg - Al LDH, where it forms H^+ and CO_3^{2-} ions by reacting with the intercalated H_2O (Eq. (3)); the formed CO_3^{2-} ions then replace the pre-intercalated CO_3^{2-} (Eq. (4)) (Ishihara et al., 2013; Sahoo et al., 2014). However, while the HCl removal rate decreases upon introducing 5 % CO_2 , it remains relatively unchanged when the concentration of CO_2 is further increased to 15 %, likely due to the limited dissolution of CO_2 in intercalated H_2O under atmospheric pressure owing to the low amount of intercalated H_2O .

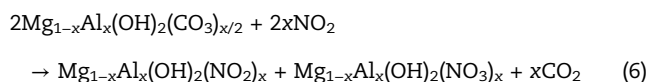


Similar to the case for HCl , coexisting CO_2 also has an inhibitory effect on SO_2 removal by CO_3 - Mg - Al LDH. SO_2

removal occurs by the reaction between intercalated CO_3^{2-} and SO_2 (chemisorption) (Eq. (5)) and SO_2 adsorption onto the Mg - Al LDH surface (physisorption) (Kameda et al., 2019b). Therefore, the decreased SO_2 removal rate in the presence of CO_2 is attributed to the exchange reaction between intercalated CO_3^{2-} and gaseous CO_2 (Eqs. (3) and (4)), which inhibits the reaction between intercalated CO_3^{2-} and gaseous SO_2 (Eq. (5)).



In contrast, while the NO_2 removal rate also decreases in the presence of coexisting CO_2 , the trend is not consistent with regard to CO_2 concentration. The removal of NO_2 is a result of both the reaction between intercalated CO_3^{2-} and NO_2 (chemisorption) (Eq. (6)) and NO_2 adsorption onto the Mg - Al LDH surface (physisorption) (Kameda et al., 2019c). In the presence of gaseous CO_2 , the exchange reaction between intercalated CO_3^{2-} and gaseous CO_2 (Eqs. (3) and (4)) also plays a role in inhibiting the reaction between intercalated CO_3^{2-} and gaseous NO_2 (Eq. (6)).



In this study, we aimed to develop a process for collecting and concentrating atmospheric CO_2 in the presence of atmospheric contaminants. We selected CO_3 - Mg - Al LDH for this purpose, because it has been shown to selectively adsorb CO_2 at a higher rate than other acid gas components. Previous studies have evaluated the adsorption properties of CO_3 - Mg - Al LDH in evacuated reaction tubes, where CO_2 is introduced and adsorption is calculated from the difference in pressure before and after adsorption (Sharma et al., 2008; Wang et al., 2012; Ramírez-Moreno et al., 2014; Li et al., 2017; Liu et al., 2020). While these studies aimed to evaluate the properties of the adsorbent, our goal was to develop a novel process for collecting and concentrating atmospheric CO_2 in the presence of atmospheric contaminants. Therefore, we conducted absorption experiments in reaction tubes filled with CO_3 - Mg - Al LDH, to which a mixture of CO_2 and N_2 gas was introduced, with the aim of collecting highly concentrated CO_2 . We evaluated the appropriate conditions for CO_2 adsorption and desorption for efficient CO_2 collection.

2. Experimental

The specific surface area and mean pore diameter of CO_3 - Mg - Al LDH (Mg/Al molar ratio = 2, 3, and 4; Sakai Chemical Industry Co., Ltd.; Fig. S1) were measured by nitrogen adsorption (BELSORP-miniX, Microtrac MRB Co., Ltd.). In addition, the CO_2 adsorption quantity was measured at a temperature of 25 °C and relative pressure of 0–1.0 using the same instrument. The CO_2 in CO_3 - Mg - Al LDH (Mg/Al = 3) was quantified using a temperature-programmed desorption (TPD) mass spectrometer (BELCAT II, Microtrac MRB). The sample was heated to 900 °C at a rate of 10 °C/min under a flow of nitrogen gas (30 mL/min), and the mass spectrum intensity was monitored at m/z = 44.

Fig. S2 shows the apparatus used for the dry CO_2 adsorption experiments. A 0.5 g sample of CO_3 - Mg - Al LDH (Mg/Al = 2, 3, and 4) was placed in a reaction tube with an internal diameter of 16 mm and fixed in place with glass wool. The

reaction tube was then inserted into a tubular electric furnace and heated to temperatures of 30, 60, 100, or 200 °C. A mixture of CO₂ (400 ppm) and N₂ (bal.) gases was then passed through the reaction tube at a linear speed of 0.500 m/min. The concentration of unreacted CO₂ gas was measured at the outlet using a gas analyzer (Model 410i, Nippon Thermo Co., Ltd.), and the CO₂ adsorption quantity was calculated by summing the CO₂ adsorption quantity per minute using Eq. (7). In this way, we evaluated the effects of the Mg/Al molar ratio in CO₃Mg–Al LDH and the reaction temperature on the CO₂ adsorption quantity.

$$\sum_{t=1}^T \left(Q \times (C_{in} - C_{out}) \times t \times \frac{1}{V_{m,CO_2}} \right) \quad (7)$$

where Q [mL/min] is the flow rate; C_{in} and C_{out} [ppm] are the CO₂ gas concentrations at the inlet and outlet, respectively; t [min] is time; and V_{m,CO_2} [mL/mmol] is the molar volume of CO₂. V_{m,CO_2} was calculated using the ideal gas law under the assumption that CO₂ behaves as an ideal gas.

Fig. S3 shows the apparatus used for dry CO₂ desorption experiments. A 0.25 g sample of CO₃Mg–Al LDH (Mg/Al = 3) that had adsorbed CO₂ at 60 °C for 250 min was placed in a reaction tube with an internal diameter of 16 mm and fixed in place with glass wool. The reaction tube was then placed in a tubular electric furnace and heated to 100, 200, or 300 °C. N₂ gas was passed through the reaction tube at linear speeds of 0.167, 0.250, or 0.500 m/min. The concentration of desorbed CO₂ was measured at the outlet using a gas analyzer, and the CO₂ desorption quantity was calculated by summing the CO₂ desorption quantity per minute using Eq. (8). In this way, we evaluated the effects of the reaction temperature and linear speed on the CO₂ desorption quantity by CO₃Mg–Al LDH.

$$\sum_{t=1}^T \left(Q \times C_{out} \times t \times \frac{1}{V_{m,CO_2}} \right) \quad (8)$$

3. Results and discussion

Table 1 presents the specific surface area and average pore size of each CO₃Mg–Al LDH sample, as determined by Brunauer–Emmett–Teller and Barrett–Joyner–Halenda analyses, respectively. The specific surface area was highest for the sample with an Mg/Al molar ratio of 4, although it did not change significantly between samples. The mean pore diameter increased with increasing Mg/Al molar ratio. Fig. 1 shows the effect of the Mg/Al molar ratio on the CO₂ adsorption isotherm. At each pressure, the CO₂ adsorption quantity was high when the Mg/Al molar ratio was 3 or 4, and low when the Mg/Al molar ratio was 2. As stated in Section 1, CO₂ dissolves in the intercalated H₂O in Mg–Al LDH. Using Eq. (1), with Mg/Al molar ratios of 2, 3, and 4, the coefficient m , which represents the quantity of intercalated H₂O, is calculated as 0.51, 0.63, and 0.70, respectively. In other words, the higher the Mg/Al molar ratio, the greater the quantity of

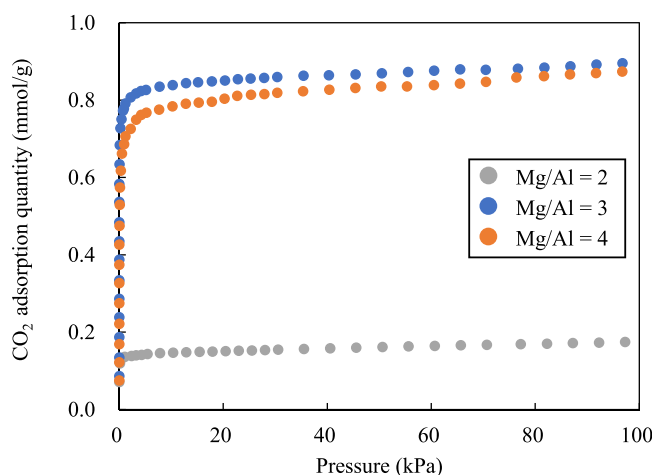


Fig. 1 – Effect of Mg/Al molar ratio on the CO₂ adsorption isotherm of CO₃Mg–Al LDH.

intercalated H₂O. This means that CO₂ would dissolve more easily in the LDH, resulting in an increased CO₂ adsorption quantity.

The CO₂ adsorption isotherm data (Fig. 1) were fitted to the Langmuir and Freundlich models. The Langmuir model (Eq. (9)) is a monolayer adsorption model that assumes adsorbate molecules and adsorption sites have a 1:1 ratio, which often applies during chemisorption.

$$V = \frac{V_m a P}{1 + a P} \quad (9)$$

where P is the equilibrium pressure; V is the equilibrium adsorption quantity; V_m is the maximum adsorption quantity; and a is the adsorption equilibrium coefficient. Modification of Eq. (9) yields Eq. (10).

$$\frac{P}{V} = \frac{1}{V_m a} + \frac{P}{V_m} \quad (10)$$

In contrast, the Freundlich model (Eq. (11)) is a multilayer adsorption model, which often applies during adsorption to heterogeneous surfaces.

$$V = K_F P^{\frac{1}{n}} \quad (11)$$

where K_F is the Freundlich coefficient and n is the constant. Elimination of the logarithms from both sides of Eq. (11) yields Eq. (12).

$$\log V = \log K_F + \frac{1}{n} \log P \quad (12)$$

From the adsorption isotherm in Fig. 1, the CO₂ adsorption quantity appears to saturate at low CO₂ concentrations. Therefore, the adsorption model was applied in the range of 0–25 kPa. Figs. S4 and S5 show the Langmuir and Freundlich plots, respectively, and Table 2 shows the R^2 values for each adsorption model, as well as the maximum CO₂ adsorption quantities calculated using the Langmuir formula. The Langmuir model had high R^2 values for all the CO₃Mg–Al LDHs, indicating that monolayer adsorption took place. The maximum CO₂ adsorption quantity calculated from the Langmuir equation was 0.794 mmol/g for CO₃Mg–Al LDH with an Mg/Al molar ratio of 3. Santamaría et al. (2023) recently reviewed the use of LDHs for CO₂ adsorption at moderate temperatures (200–450 °C). Although the adsorption temperatures were higher than those used here, the CO₂ adsorption capacities of the LDHs they reviewed were similar to our results.

Table 1 – Specific surface area and mean pore diameter of CO₃Mg–Al LDH.

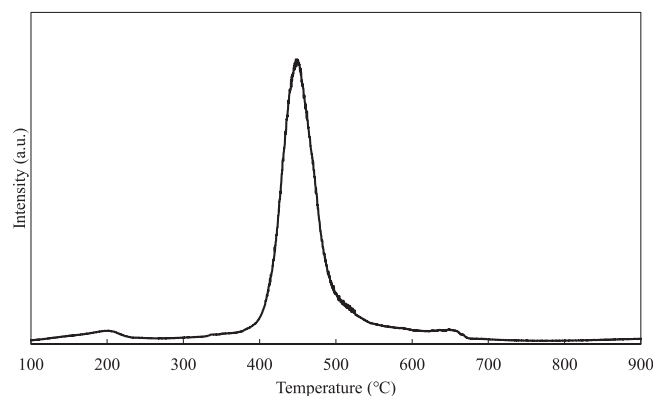
Mg/Al molar ratio	Specific surface area (m ² /g)	Average pore size (nm)
2	10.6	49.8
3	16.3	73.0
4	16.6	106.7

Table 2 – R^2 values for Langmuir and Freundlich models and maximum CO_2 adsorption quantities as calculated by Langmuir formula.

Mg/Al molar ratio	R^2 (Langmuir)	R^2 (Freundlich)	Calculated maximum CO_2 adsorption quantity (mmol/g)
2	0.99	0.74	0.139
3	0.99	0.61	0.794
4	0.99	0.67	0.751

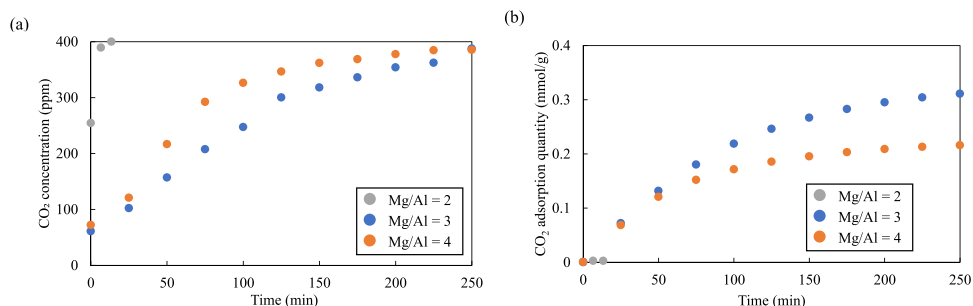
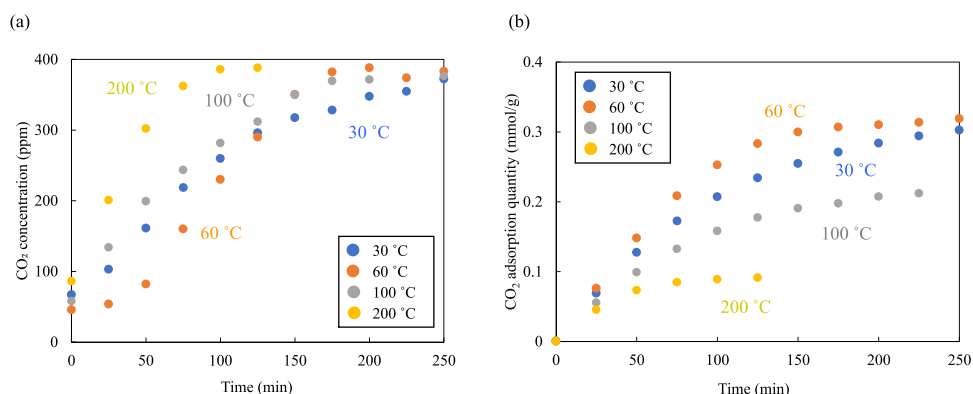
The effects of the Mg/Al molar ratio on the CO_2 concentration in the outlet gas and the CO_2 adsorption quantity at 30 °C are shown in Fig. 2(a) and (b), respectively. For the samples with Mg/Al molar ratios of 3 and 4, the CO_2 concentration at the outlet approached the inlet concentration (i.e., 400 ppm) after approximately 250 min, confirming that breakthrough occurred (that is, the adsorbent was no longer effectively capturing CO_2). For the sample with an Mg/Al molar ratio of 2, this value was reached much faster (within 20 min). In addition, the CO_2 adsorption quantity gradually approached a constant value (Fig. 2(b)). The highest CO_2 adsorption quantity of 0.310 mmol/g was measured for the sample with an Mg/Al molar ratio of 3. This finding is consistent with the trend in CO_2 adsorption quantity calculated using the Langmuir formula, as shown in Table 2.

Fig. 3(a) and (b) show the effect of temperature on the CO_2 concentration in the outlet gas and the amount of CO_2 adsorbed by $\text{CO}_3\cdot\text{Mg-Al LDH}$ (Mg/Al = 3), respectively. Irrespective of temperature, the CO_2 concentration in the outlet gas approached the initial concentration (i.e., 400 ppm) after 100–250 min, indicating that breakthrough occurred. In addition, the CO_2 adsorption quantity gradually approached a

**Fig. 4 – CO_2 -TPD spectrum of $\text{CO}_3\cdot\text{Mg-Al LDH}$ (Mg/Al = 3).**

constant value. The maximum CO_2 adsorption quantity of 0.319 mmol/g was achieved at 60 °C. Because CO_2 dissolves in the intercalated H_2O in Mg–Al LDH, the quantity of intercalated H_2O would be lower at 100 and 200 °C due to the release of intercalated H_2O , which starts at 100 °C. Therefore, CO_2 dissolves less readily. In contrast, at 30 and 60 °C, the presence of intercalated H_2O in Mg–Al LDH allows for the dissolution of CO_2 and increases the adsorption capacity. In addition, as shown in Table 2, CO_2 adsorption by $\text{CO}_3\cdot\text{Mg-Al LDH}$ occurs by monolayer adsorption, which is a form of chemisorption. Therefore, the reaction is promoted by an increase in temperature from 30° to 60 °C, resulting in the maximum CO_2 adsorption quantity being observed at 60 °C.

Fig. 4 shows the CO_2 -TPD spectrum of $\text{CO}_3\cdot\text{Mg-Al LDH}$ (Mg/Al = 3). There are three peaks of CO_2 release, which indicate that the CO_2 dissolved in the intercalated H_2O is released at approximately 200 °C, while the intercalated CO_3^{2-} is released in two stages at 350–600 and 650–700 °C. Thus, the

**Fig. 2 – Effect of Mg/Al molar ratio of $\text{CO}_3\cdot\text{Mg-Al LDH}$ on (a) CO_2 concentration in outlet gas and (b) quantity of CO_2 adsorbed at 30 °C.****Fig. 3 – Effect of temperature on (a) CO_2 concentration in outlet gas and (b) quantity of CO_2 adsorbed using $\text{CO}_3\cdot\text{Mg-Al LDH}$ (Mg/Al = 3).**

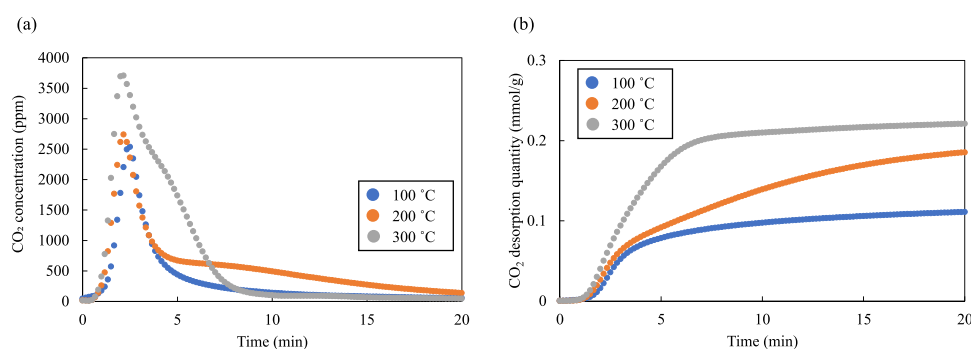


Fig. 5 – Effect of temperature on (a) CO₂ concentration in outlet gas and (b) quantity of CO₂ desorbed from CO₂-adsorbed CO₃-Mg-Al LDH at a linear speed of 0.500 m/min.

Table 3 – Effect of temperature on the maximum CO₂ concentration at the outlet and CO₂ desorption quantity (linear speed = 0.500 m/min).

Temperature (°C)	Maximum CO ₂ concentration (ppm)	Quantity of desorbed CO ₂ (mmol/g)
100	2536	0.111
200	2737	0.185
300	3707	0.221

Table 4 – Effect of linear speed on the maximum CO₂ concentration at the outlet and CO₂ desorption quantity (temperature = 300 °C).

Linear speed (m/min)	Maximum CO ₂ concentration (ppm)	Quantity of desorbed CO ₂ (mmol/g)
0.167	2454	0.170
0.250	6232	0.160
0.500	3707	0.221

temperature for CO₂ desorption was limited to 300 °C to ensure the CO₂ dissolved in the intercalated H₂O is released while the intercalated CO₃²⁻ is retained. Fig. 5(a) and (b) show the effect of temperature on the CO₂ concentration in the outlet gas and the amount of CO₂ desorbed from CO₂-adsorbed CO₃-Mg-Al LDH at a linear speed of 0.500 m/min, respectively. Additionally, Table 3 lists the maximum CO₂ concentrations at the outlet and the CO₂ desorption quantities at each temperature. Regardless of the temperature, the CO₂ concentration at the outlet increased rapidly due to CO₂ desorption, and then asymptotically approached 0 ppm within 20 min, suggesting that all the CO₂ that could be released was released. The CO₂ concentration and CO₂ desorption quantity both reached maximum values at 300 °C, confirming that CO₂ is more readily released at higher temperatures.

Fig. 6(a) and (b) show the effect of linear speed on the CO₂ concentration in the outlet gas and the CO₂ desorption quantity from CO₂-adsorbed CO₃-Mg-Al LDH at 300 °C, respectively. In addition, Table 4 shows the effects of linear speed on the maximum CO₂ concentration at the outlet and

the CO₂ desorption quantity. Regardless of the linear speed, the CO₂ concentration at the outlet increased rapidly due to CO₂ desorption, after which it asymptotically approached 0 ppm within 20 min, indicating that all the CO₂ that could be released was released. At a linear speed of 0.250 m/min, the maximum CO₂ concentration at the outlet was 6232 ppm, representing a 15.6-fold increase from the initial concentration of 400 ppm. In relation to the CO₂ concentration at the outlet asymptotically approaching 0 ppm, the CO₂ collection quantity asymptotically approached a constant value, and the maximum value of 0.221 mmol/g was achieved when the linear speed was 0.500 m/min.

4. Conclusions

The results of this study indicate that the CO₂ adsorption capacity of CO₃-Mg-Al LDH is highest when the Mg/Al molar ratio is 3 or 4. This was ascribed to the greater quantity of intercalated H₂O at higher Mg/Al ratios, which allows for better CO₂ dissolution. Using the Langmuir model, it was determined that monolayer adsorption occurs for all

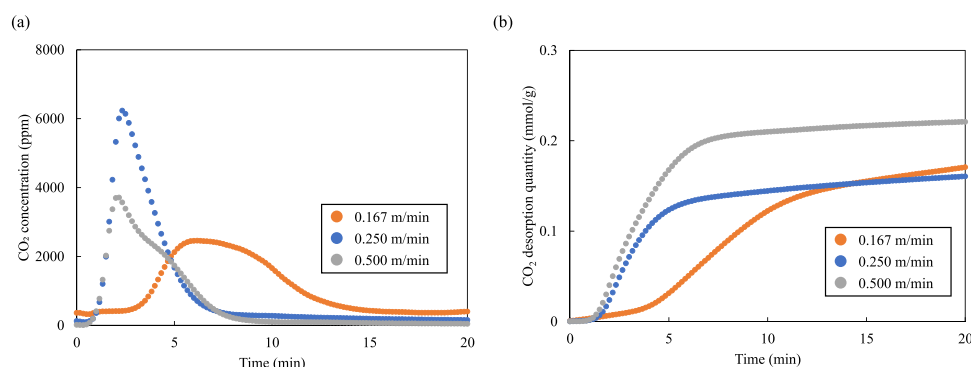


Fig. 6 – Effect of linear speed on (a) CO₂ concentration in outlet gas and (b) quantity of CO₂ desorbed from CO₂-adsorbed CO₃-Mg-Al LDH at a temperature of 300 °C.

samples, regardless of the Mg/Al molar ratio. The maximum CO₂ adsorption quantity calculated using the Langmuir formula was 0.794 mmol/g for the CO₃-Mg–Al LDH with an Mg/Al molar ratio of 3.

In dry CO₂ adsorption experiments, the maximum CO₂ adsorption quantity was 0.319 mmol/g when the Mg/Al molar ratio was 3 and the temperature was 60 °C. This was ascribed to the fact that at higher temperatures (100 and 200 °C), the release of intercalated H₂O leads to decreased CO₂ dissolution, while at lower temperatures (30 and 60 °C), the presence of intercalated H₂O allows for greater CO₂ dissolution and adsorption. Additionally, as the temperature increases from 30° to 60°C, the reaction is promoted; thus, the maximum absorption quantity was observed at 60 °C.

In dry CO₂ desorption experiments, the maximum CO₂ desorption temperature was set to 300 °C, at which point the CO₂ dissolved in intercalated H₂O in Mg–Al LDH was released, but the intercalated CO₃²⁻ in Mg–Al LDH was not. The maximum CO₂ concentration and desorption quantity were observed at 300 °C, indicating that CO₂ release increases with temperature. At 300 °C and a linear speed of 0.250 m/min, the maximum CO₂ concentration reached 6232 ppm, a 15.6-fold increase from the initial concentration of 400 ppm. The highest CO₂ collection quantity of 0.221 mmol/g was achieved when the linear speed was 0.500 m/min.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cherd.2023.04.065](https://doi.org/10.1016/j.cherd.2023.04.065).

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